

T-62 – DESIGN AND CONSTRUCTION OF A REVERSIBLE 500G COMBAT ROBOT WITH MODULAR ROTARY WEAPON

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***ABSTRACT:** This paper describes the entire design process and build documentation of our reversible 500 gram combat robot, T-62, competing in the category “cage wrestling 500g” at the Minoan Robotports Competition Global Olympiad 2026. It was built with design references from our prior competition robot T-34, which competed at Robochallenge 2025. We identified weaknesses from Robochallenge to address during design: a prone-to-break gear-driven weapon transmission, exposed wiring connections which went loose upon impact and a lack of replaceable subsystems which caused slow field repairs. Improvements implemented in T-62 include a lower profile chassis with 75 degree angled front armor plate, GT2 belt-driven rotary weapon actuation, a custom PCB to consolidate and securely fasten all electronics, and modular mechanical design such that any crucial component can be swapped out in 5-10 minutes.*

***KEYWORDS:** combat robot, reversible design, rotary weapon, modular chassis, ESP32*

1. Introduction

Combat robotics is an emerging branch of engineering where teams design, fabricate, and remotely control robots to fight against each other inside of an enclosed cage. Many attributes of a competition determine the design of these robots; however, robotics combat is one of few competitions that challenge mechanical strength, electrical survivability, and software dependability all at once. The weight class of 500 gram (1 lb) has become one of the largest following categories of competitive robotics because teams can build and iterate designs quickly with minimal material costs and attend multiple competitions each season as students. T-62 was designed to fit into the “cage wrestling 500g” weight class of robots to be used in the Minoan Robotports Competition Global Olympiad 2026.

The improvements over Robochallenge 2025's T-34 come from experiences operating that robot and were used to motivate each design change. This project's goals are to increase overall weapon system reliability, minimize downtime during competitions to perform repairs, increase survivability from high-energy hits, and fit into the 500 g weight-class category. This document walks through the design process of each of the robots' subsystems including mechanical design and chassis design, weapon drive design, materials and manufacturing, electrical design, and firmware. Each section details the reasoning behind each design decision as well as trade studies that were performed and what we learned from T-34.

2. State of the art

The 500 g rulesets are fairly consistent across international borders, typically requiring a maximum weight, maximum height and width, weapon safety such as a kill switch and keeping robots confined to the arena. Because of this, teams use different tactics when designing their robots.

Some common weapon types in this weight class are horizontal bar spinners, vertical disc spinners, drum spinners, and body spinners. Each of these weapon types have varying levels of potential energy, durability and fit many types of frames.

One recent trend in the smaller size classes are reversible robots, or robots capable of fighting both right-side up and upside down correctly. This removes the weight and complexity of adding a mechanism to change the orientation yourself but does come with some restrictions such as weapon length and symmetry of the frame. Example of a reversible bot can be found in Figure 1 below.

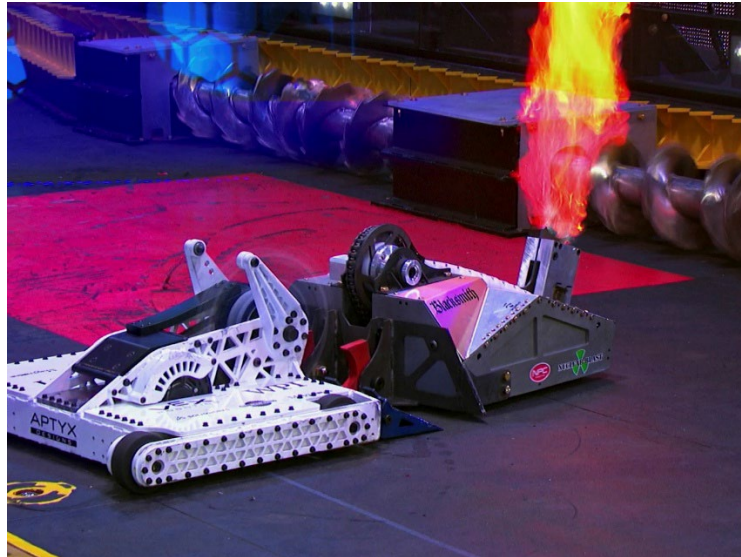


Fig. 1. The robot seen on the left is side was of great inspiration for our project [1]

3. Project development (materials and methods):

The teams first robot, T-34 competed in Robochallenge 2025, being able to gather information during an actual competition environment. The team discovered three types of failures from tournament play. First, backlash had built up rapidly in the gear driven weapon transmission as a result of repeated impact loads. The robot would become less effective as the match wore on until the gear mesh would simply bind up from physical interference. Second, the gear system provided unnecessary weight vs the expected weapon performance provided.

Second, wiring was run internally and never mechanically secured causing vibrations when struck to cause disconnects during matches. Last, field reparability was poor; changing out a drive motor or the battery meant removing several structural plates. Repairs were taking 20-30 minutes when only 15 were allowed between matches. When building T-62 these failures were kept in mind. Keeping a low center of mass was prioritized by having the chassis sit as low to the ground as possible and by placing the battery and heaviest electrical components towards the bottom of the robot while inside the lowest internal volume of the chassis.

Its ability to be completely reversible is a massive advantage, all drive and weapon movements operate as intended no matter if the robot flips over during combat and lands either upside down or right side up. Eliminating the need for a self-righting mechanism as well as reducing the number of total parts. The front plate angles forward at 75 degrees above the horizontal plane, giving the robot a wedge shape to help lift incoming horizontal spinning projectiles as well reducing the gap between the front plate and ground.

Weapon system holds multiple plates mounted on a over-dimensioned steel axle. The axle stock is heavy gauge steel and the weapon is driven by a dedicated brushless motor connected through a timing belt transmission, through a GT2 profile belt. The larger axle has two purposes. The larger the axle the more rotational inertia the weapon will have allowing for more kinetic energy stored at the same angular velocity and also the thicker axle resists bending and shear loads better when struck away from center.

Material distribution was decided based on the mechanical function of each structural zone as well as its exposure to impact loading. Non-load bearing interior regions of the chassis were printed using PLA-LW (lightweight PLA), a foamed extrusion filament with an internal microcellular structure which brings its bulk density down to about 0.5-0.7 g/cm³ [2], approximately 40-50% lower than normal PLA [3]. This allowed for dramatic mass savings in structural zones that did not require high performance. Zones that experience higher mechanical loads (weapon mount, motor mounts, hinge locations, etc.) were printed using XPETG or PETG-CF (carbon-fiber reinforced PETG) for increased stiffness compared to normal PETG while still maintaining its high toughness and impact resistance. High-impact risk exterior armor surfaces were supplemented with 1 mm aluminum sheet to provide a rigid metal surface that can spread contact energy across a wider zone instead of being applied to a single point in the polymer. Material distribution is summarized in Table 1.

Our entire design was modeled using Autodesk Fusion 360 which allows for built in mass properties analysis, interference checks, and parametric editing. Our current CAD model comes in at ~400g without assembly fasteners; we left our finalized design under the 500g category limit by ~20g to allow for field reinforcement of weak spots found during testing. The robot actually weighs 492g so we were within our margin of error.

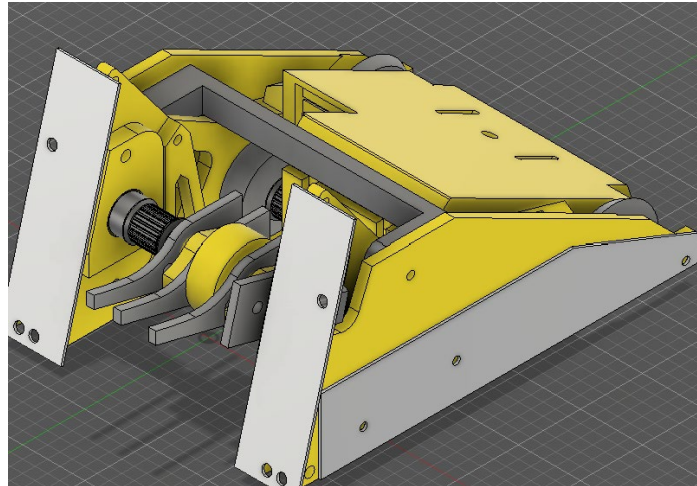


Fig. 2. The robot as seen from Fusion 360 with all the attached external armor

Table 1. Materials Used

Structural Zone	Material	Key Property
General chassis, interior panels	PLA-LW	Low density (~ 0.6 g/cm ³), adequate rigidity
Weapon mount, motor brackets	XPETG / PETG-CF	High toughness, impact resistance, stiffness
External armor surfaces	1 mm Aluminum	Hard surface, energy distribution on impact
Prototype validation parts	PLA (standard)	Low cost, rapid print iteration

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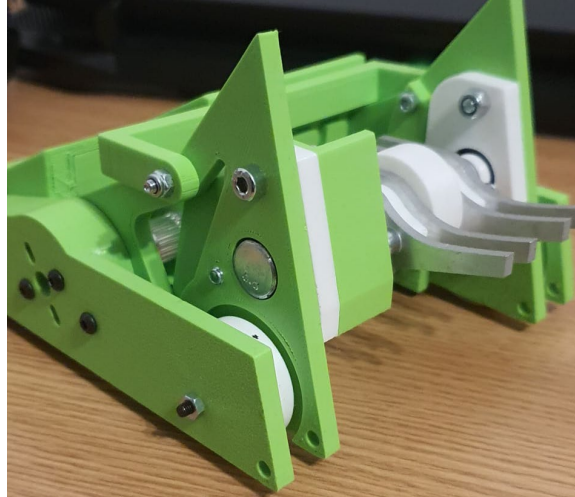


Fig. 2 the first prototype fully assembled from spare PLA

All electronic components are integrated on a single custom PCB, designed in EasyEDA. This consolidates all connections onto soldered traces and footprints rather than hand-wired point-to-point connections, directly eliminating the cable-disconnection failure mode of T-34. The complete electrical architecture of T-62 is presented in Figure 3 with the PCB design in Figure 2.

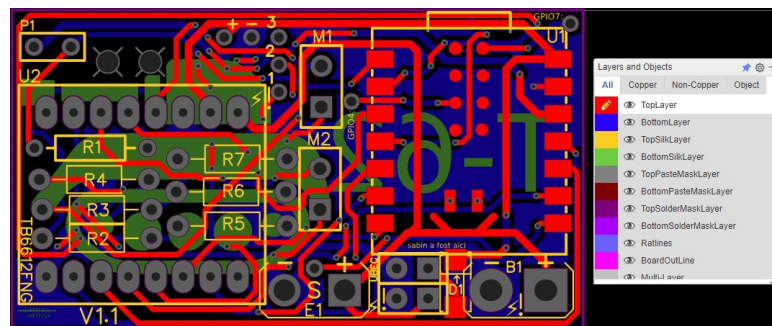


Fig. 3 the PCB layout, from EasyEDA

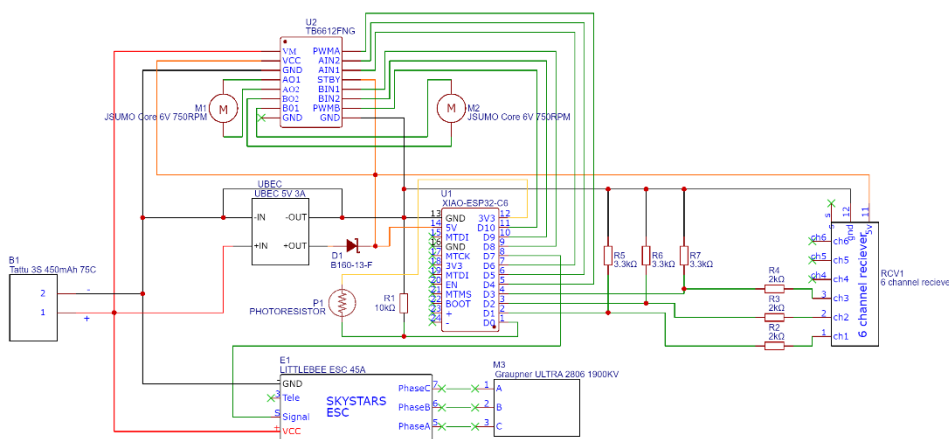


Fig. 4 T-62 electrical schematic

Primary power comes from a Tattu 3S 450 mAh 75C LiPo battery (nominal voltage: 11.1 V). Power from the battery goes through a Universal Battery Eliminator Circuit, UBEC, followed by a 5V 3A linear regulator to provide power to the microcontroller and receiver circuitry at a constant 5 volts regardless of battery voltage due to discharge. There is a protection

diode (B160-13-F) in series with the output of the UBEC. The main microcontroller board used is SeeedStudio's XIAO ESP32-C6, as it has a small footprint (21×17.5 mm), had enough GPIO pins for all needed IO functions and can work with the Arduino IDE C++ program environment. Drive motors M1 and M2 (JSUMO Core 6V 750 RPM brushed DC motors) are connected to the TB6612FNG dual H-bridge driver IC (U2) which is directly driven by PWM and direction pins from the ESP32-C6.

The weapon brushless motor, M3 (Graupner ULTRA 2806, 1900 KV) is connected to a LITTLEBEE 45A ESC (E1) which accepts its PWM setpoint from the microcontroller. Received RC input was originally done with a 6 channel receiver (RCV1), but in the final build it was replaced by the FS2A 4 channel receiver because of size and weight; 3 channels are used (left-right, forward-back, weapon) which are read through a voltage divider (R2–R7, 2 k Ω /3.3 k Ω) to shift the receiver's 5V signal levels down to the ESP32-C6's GPIO input voltage of 3.3V.

Table 2. A list of the electronic components mentioned above

Reference	Component	Specification
U1	Microcontroller	Seeed XIAO ESP32-C6, 3.3 V, 160 MHz
U2	Motor driver	TB6612FNG, 13.5 V max, 1.2 A cont. / 3.2 A peak per channel
E1	Weapon ESC	LITTLEBEE 45A, BLHeli_S
M1, M2	Drive motors	JSUMO Core 6V 750 RPM, brushed DC
M3	Weapon motor	Graupner ULTRA 2806, 1900 KV, brushless
B1	Battery	Tattu 3S 450 mAh 75C LiPo, 11.1 V nominal
UBEC	Voltage regulator	UBEC 5V 3A
D1	Protection diode	B160-13-F, Schottky
RCV1	RC Receiver	4-channel PWM receiver
R2, R3	Voltage divider (low)	2 k Ω , signal level shifting
R4, R5, R6, R7	Voltage divider (high)	3.3 k Ω , signal level shifting
P1, R1	Light sensor circuit	Photoresistor + 10 k Ω reference

The drive control algorithm converts two RC receiver channels: forward/backward (FB) and left/right (LR), to independent speed control commands for the left and right drive motors based on a differential (tank) mixing model. Receiver pulse width values are normalized to the receiver neutral pulse value of 1500 μ s. Processed pulse width values, described in equations (3) and (4), are mapped to integer motor speed commands ranging from $[-127, +127]$ using the Arduino map() function:

The computed throttle value T and steering value S are combined as follows to yield the left motor command M_L and right motor command M_R , where constrain() forces the output to the valid range of the TB6612FNG PWM interface. Straight drive with no steering ($S = 0$) yields equal power to both motors. Pure rotation ($T = 0$) causes one motor to drive forward and the other to reverse at the same speed.

$$M_L = \text{constrain}(T + S, -127, 127) \quad (1)$$

$$M_R = \text{constrain}(T - S, -127, 127) \quad (2)$$

Each raw pulse width sample passes through a pipeline of three signal filters before producing the RC signal output. First is a 7-sample median filter to reject impulse noise:

$$y_{\text{median}}[n] = \text{median}(x[n], x[n - 1], \dots, x[n - 6]) \quad (3)$$

Next is a spike filter that eliminates transitory outliers by discarding new input values until they have been repeated for at least 2 frames. Finally, an Exponential Moving Average (EMA) is used to smooth accepted values:

$$y_{\text{EMA}}[n] = \alpha \cdot y_{\text{med}}[n] + (1 - \alpha) \cdot y_{\text{EMA}}[n - 1] \quad (4)$$

Where $\alpha = 0.75$ is the smoothing factor coded into the firmware. Larger α values bias towards responsiveness at the cost of smoothness; this specific value was determined through trial-and-error bench tuning. If we examine the kinetic energy stored behind the operating speed of the spinning weapon we get:

$$E_k = 1/2 \cdot I \cdot \omega^2 J \quad (5)$$

Where $I [kg \times m^2]$ is the weapons moment of inertia around the axis of rotation and $\omega [\text{rad/s}]$ is the angular velocity at operating speed. Increasing I by weapon disc geometry and axle weight was an intentional design choice, as more stored kinetic energy means more energy available for translation into blunt force onto the opponent.

4. Development and results:

T-62 participated in the Minoan RobotSports Competition Global Olympiad 2026 in the division “cage wrestling 500g”. T-62's opponent was a carbon fiber horizontal bar spinner who would eventually go on to place 1st overall. The opponent allocated most of its 500g weight budget into a spinning metal bar to store as much kinetic energy as possible. This created one of if not the toughest opponent any polymer robot in this weight class will face. The amount of energy a metal horizontal spinner can output per strike is greater than what nearly every other material class can absorb.

Large visible cracks developed on the T-62's PLA-LW chassis plates due to fast violent shearing from repeated high magnitude impacts. Robot's aluminum plating buckled out and many of the M3 screws deformed past point of use at connection points. Each of these failure modes provided helpful information on what parts of the opponent to focus on going forward whether in defeat or victory. Luckily all electronics survived the match unscathed, including custom PCB, XIAO ESP32-C6, TB6612FNG, and weapon ESC. The weapon could still be actuated to spin up post-match and both drive motors responded to directional input even after chassis was torn from them.

This demonstrates that our current PCB integration scheme is an improvement over the wiring malfunctions present in T-34 and that our next most probable failures will be within the framing and armor aspects of the build. A few takeaways from an engineering standpoint are that PLA-LW simply cannot handle direct high force-rate applications from horizontal spinners and M3 screws suck at maintaining joint strength when under stress bent.

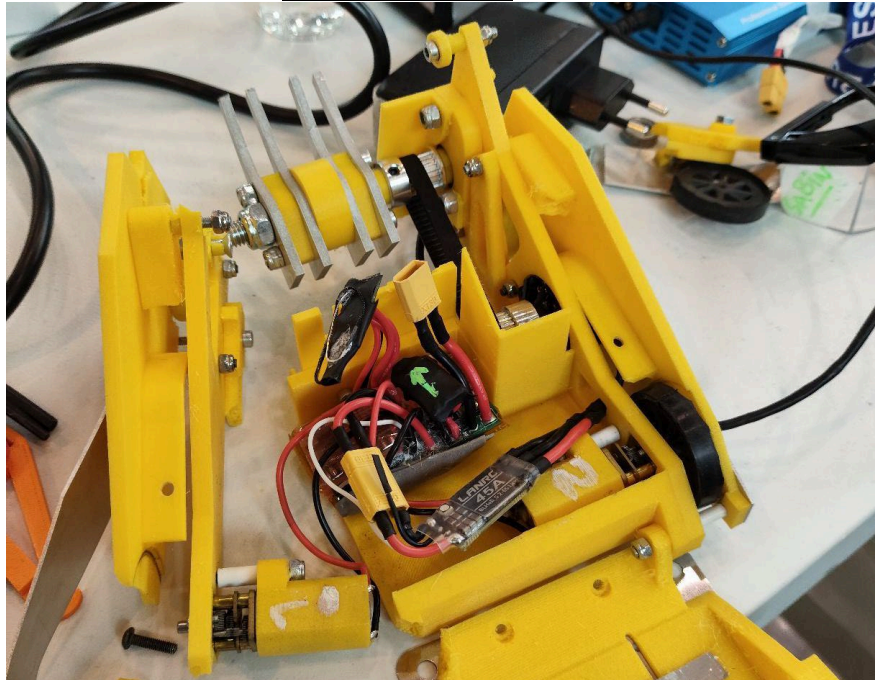


Fig. 5 image showing the catastrophic damage suffered after MRC



Fig. 6 image showing the finished T-62 robot (and its weight)

5. Conclusions

This paper encompassed the mechanical design and fabrication of T-62, a reversible 500g competitive fighting robot designed by a group of students. By developing the machine three key innovations were observed, when compared to the previous iteration, T-34.

The modular mechanical design allows all mechanical subsystems to be swapped out in under 10 minutes. Solving the problem of time spent repairing T-34 during matches as discovered at Robochallenge 2025 [4]. The timing belt based weapon transmission replaced the

weak geared drivetrain design, removing backlash and decreasing weight of system. The lightweight, custom PCB based electronics were validated at the Minoan Robotsports Competition Global Olympiad with not a single wire parted, even after severe damage to the chassis (see Figure 5).

Improvements to be considered are swapping out PLA-LW for a material with higher toughness in areas of direct impact, revising M3 screw joints to avoid shock bent failure and developing the capability to display battery voltage and motor current in real time with the help of ESP32-C6s onboard Wi-Fi wireless telemetry.

6. Bibliografy

- [1]. *** BattleBots season 3; Prime Video
- [2]. *** Polymaker Wiki - PLA-LW Technical Data Sheet
- [3]. *** Polymaker Wiki - PolyLite™ PLA Technical Data Sheet
- [4]. *** Robochallenge 2025 — Official Rulebook, Category 500g

9. Notations

The following symbols are used throughout the paper:

T = normalized throttle input from RC receiver (forward/backward channel)
[dimensionless, range -127 to $+127$ after mapping]

S = normalized steering input from RC receiver (left/right channel) [dimensionless, range -90 to $+90$ after mapping]

M_L = left drive motor speed command [dimensionless, -127 to $+127$]

M_R = right drive motor speed command [dimensionless, -127 to $+127$]

EMA = Exponential Moving Average

α = EMA smoothing coefficient [dimensionless]

E_k = kinetic energy of rotating weapon disc [J]

I = moment of inertia of weapon disc assembly about rotation axis [$\text{kg} \cdot \text{m}^2$]

ω = angular velocity of weapon disc [rad/s]

PWM = Pulse Width Modulation pulse width [μs]

ESC = Electronic Speed Controller

PCB = Printed Circuit Board

FDM = Fused Deposition Modeling

EMA = Exponential Moving Average

RC = Radio Control

UBEC = Universal Battery Eliminator Circuit